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BIT 2223 Mobile Computing

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 3

Wireless Middleware

Objectives

Upon completing this topic, you should be able to:

- Define wireless middle-ware
- Explain how wireless middle-ware works
- Give the functions of middle-ware
- Explain what IEEE 802.11 Series defines
- Explain specific characteristics of IEEE 802. 11 a/g, n, e and Blue-tooth.





3.1. Wireless Middleware

Wireless middleware is software that insulates applications from the underlying wireless network, making it easier to develop new wireless applications, as well as to port existing applications to the wireless environment.

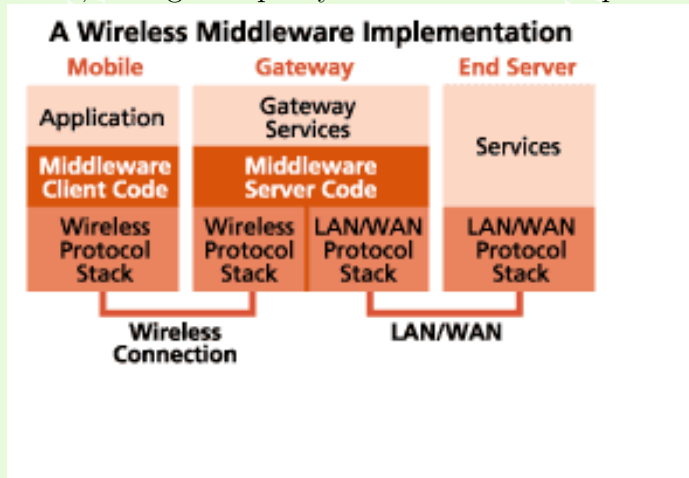
3.1.1. How it Works

Wireless middleware usually consists of client and server software. The client portion resides on the mobile computer and accepts messages from applications on the mobile computer. It re-formats these messages and forwards them across the wireless network using application-layer protocols optimized for wireless communications. The messages reach the middleware server, which typically resides on the destination LAN. The middle-



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ware server functions as a gateway to other servers and hosts on the LAN, acting as a proxy for the mobile computer.



Middleware performs the following functions, though specific details will vary depending on the actual middleware.



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- Isolates the application from connectivity issues such as intermittent connections and varying throughput.
- Minimizes the amount of data sent over the wireless connection.
- Reduces the number of back-and-forth messages required to complete a transaction.
- Queues messages when a connection is not available.
- Provides a consistent API regardless of the underlying network.

Some wireless middleware products come as toolkits with which you can develop customized wireless applications. Others work in conjunction with existing applications to make these applications effective both from a cost and performance perspective.



3.2. IEEE 802.11 Wireless LAN Standard

In 1997, the IEEE adopted the first wireless local area network (WLAN) standard, IEEE 802.11. This standard defines the media access control (MAC) and physical (PHY) layers for a LAN with wireless connectivity.

Home and business networkers looking to buy wireless local area network (WLAN) gear face an array of choices. Many products conform to the 802.11a, 802.11b, 802.11g, or 802.11n wireless standards collectively known as Wi-Fi technologies. Additionally, Bluetooth and various other non Wi-Fi technologies also exist, each are also designed for specific networking applications



3.2.1. 802.11

In 1997, the Institute of Electrical and Electronics Engineers (IEEE) created the first WLAN standard. They called it 802.11 after the name of the group formed to oversee its development. Unfortunately, 802.11 only supported a maximum network bandwidth of 2 Mbps - too slow for most applications. For this reason, ordinary 802.11 wireless products are no longer manufactured.

- **802.11 a**

802.11a supports bandwidth up to 54 Mbps and signals in a regulated frequency spectrum around 5 GHz. This higher frequency compared to 802.11b shortens the range of 802.11a networks. The higher frequency also means 802.11a signals have more difficulty penetrating walls and other obstructions.



- 802.11b

IEEE expanded on the original 802.11 standard in July 1999, creating the 802.11b specification. 802.11b supports bandwidth up to 11 Mbps, comparable to traditional Ethernet. 802.11b uses the same unregulated radio signaling frequency (2.4 GHz) as the original 802.11 standard which lowers production cost.

Pros - lowest cost; signal range is good and not easily obstructed

Cons - slowest maximum speed; home appliances may interfere on the unregulated frequency band.

- 802.11g

In 2002 and 2003, WLAN products supporting a newer standard called 802.11g emerged on the market. 802.11g attempts to



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combine the best of both 802.11a and 802.11b. 802.11g supports bandwidth up to 54 Mbps, and it uses the 2.4 Ghz frequency for greater range. 802.11g is backwards compatible with 802.11b, meaning that 802.11g access points will work with 802.11b wireless network adapters and vice versa.

- Pros of 802.11g - fast maximum speed; signal range is good and not easily obstructed
- Cons of 802.11g - costs more than 802.11b; appliances may interfere on the unregulated signal frequency
- **802.11n**

It was designed to improve on 802.11g in the amount of bandwidth supported by utilizing multiple wireless signals and antennas (called MIMO technology) instead of one.



When this standard is finalized, 802.11n connections should support data rates of over 100 Mbps. 802.11n also offers somewhat better range over earlier Wi-Fi standards due to its increased signal intensity. 802.11n equipment will be backward compatible with 802.11g gear.

3.2.2. IEEE 802.11 SERIES - Protocols for wireless networks

A protocol is a set of rules or agreed upon guidelines for communication. When communicating it is important to agree on how to do so.

On the Internet, the set of communications protocols used is called TCP/IP. TCP/IP is a collection of various protocols that each have their own special function. These protocols have



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been established by international standards bodies and are used in almost all platforms and around the globe to ensure that all devices on the Internet can communicate successfully.

- **IEEE 802.11b (Old – 1990s)**

The most prevalent is 802.11b. Equipment using 802.11b is comparatively inexpensive. The 802.11b wireless communication standard operates in the unregulated 2.4 GHz (80 MHz) frequency range. Unfortunately, so do many other devices such as cordless phones and baby monitors which can interfere with your wireless network traffic. The maximum speed for 802.11b communications is 11 mbps and approx. 500 ft range.

802.11a/g (Middle Age – mid-late 1990s)

- Standard for 5GHz NII band (300 MHz)



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- OFDM in 20 MHz with adaptive rate/codes
- Speeds of 54 Mbps, approx. 100-200 ft range

802.11n (Standard close to finalization)

- Standard in 2.4 GHz and 5 GHz band
- Adaptive OFDM /MIMO in 20/40 MHz (2-4 antennas)
- Speeds up to 600Mbps, approx. 200 ft range
- Other advances in packets utilization, antenna use, etc.

WiMAX: 802.16e:

- Wireless OFDM/MIMO standard similar to LTE cellular
- Operates in 2.5 and 3.5 MHz bands: BW is 3.5-10 MHz
- Fixed: 75 Mbps in 10-20 MHz, up to 50 mile cell radius



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- Mobile: 15 Mbps max, up to 1-2 mile cell radius
- Supports mobility and hand-off
- Controlled access: Has less interference
- Higher power: significantly higher range

Wi-Fi: 802.11n

- Optimized for high speed local access (300 Mbps in 40 MHz: 2x2)
- Unlicensed: Two bands (2.4/5) that are uniform world-wide
- No hand-off or mobility support
- Random access
- Low power: range obtained via relays or mesh



Blue-tooth

Blue-tooth devices transmit at relatively low power and have a range of only 30 feet or so. Blue-tooth networks also use the unregulated 2.4 Ghz frequency range and are limited to a maximum of eight connected devices. The maximum transmission speed only goes to 1 mbps. There are many other standards being developed and introduced in this wireless networking field.

IEEE 802.11 Series Summary

Within the IEEE 802.11 Working Group, the following IEEE Standards Association Standard and Amendments exist:

- IEEE 802.11 - The WLAN standard was original 1 Mbit/s and 2 Mbit/s, 2.4 GHz RF and infrared [IR] standard (1997), all the others listed below are Amendments to this



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standard, except for Recommended Practices 802.11F and 802.11T.

- IEEE 802.11a - 54 Mbit/s, 5 GHz standard (1999, shipping products in 2001)
- IEEE 802.11b - Enhancements to 802.11 to support 5.5 and 11 Mbit/s (1999)
- IEEE 802.11c - Bridge operation procedures; included in the IEEE 802.1D standard (2001)
- IEEE 802.11d - International (country-to-country) roaming extensions (2001)
- IEEE 802.11e - Enhancements: QoS, including packet bursting (2005)
- IEEE 802.11F - Inter-Access Point Protocol (2003) With-



drawn February 2006

- IEEE 802.11g - 54 Mbit/s, 2.4 GHz standard (backwards compatible with b) (2003)
- IEEE 802.11h - Spectrum Managed 802.11a (5 GHz) for European compatibility (2004)
- IEEE 802.11i - Enhanced security (2004)
- IEEE 802.11j - Extensions for Japan (2004)
- IEEE 802.11-2007 - A new release of the standard that includes amendments a, b, d, e, g, h, i & j. (July 2007)
- IEEE 802.11k - Radio resource measurement enhancements (2008)
- IEEE 802.11n - Higher throughput improvements using



MIMO (multiple input, multiple output antennas) (November 2009)

- IEEE 802.11p - WAVE - Wireless Access for the Vehicular Environment (such as ambulances and passenger cars) (working - June 2010)
- IEEE 802.11r - Fast roaming Working "Task Group r" - (2008)
- IEEE 802.11s - Mesh Networking, Extended Service Set (ESS) (working - September 2010)
- IEEE 802.11T - Wireless Performance Prediction (WPP) - test methods and metrics recommendation cancelled
- IEEE 802.11u - Inter-working with non-802 networks (for example, cellular) (working - September 2010)



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- IEEE 802.11v - Wireless network management (working - June 2010)
- IEEE 802.11w - Protected Management Frames (working - September 2009)
- IEEE 802.11y - 3650-3700 MHz Operation in the U.S. (2008)
- IEEE 802.11z - Extensions to Direct Link Setup (DLS) (August 2007 - December 2011)
- IEEE 802.11aa - Robust streaming of Audio Video Transport Streams (March 2008 - June 2011)
- IEEE 802.11mb - Maintenance of the standard. Expected to become 802.11-2011. (ongoing)
- IEEE 802.11ac - Very High Throughput <6GHz (Septem-



ber 2008 - December 2012)

- IEEE 802.11ad - Extremely High Throughput 60GHz (December 2008 - December 2012)

3.3. Wireless transmissions

3.3.1. Antennae

An antenna is an electrical conductor or system of conductors either for radiating electromagnetic energy into the space or collecting energy from space.

- *Transmission* - radiates electromagnetic energy into space. For transmission of signal electromagnetic electrical signal from transmitter is converted into electromagnetic energy by the antennae and radiated into the surrounding atmo-



sphere.

- *Reception* - collects electromagnetic energy from space. Electromagnetic energy impinging on the antennae is converted into electrical energy and fed into receiver.
- In two-way communication, the same antenna can be used for transmission and reception
- **Types of antennae**
 - **Isotropic reflective antennae**-is a point in space which radiates power in all directions equally. The actual radiation pattern for isotropic antennae is a sphere with antennae at the center.
 - **Parabolic reflective antennae** - Mostly used by terrestrial microwave and satellite.



- **Dipole antennas:** - Half-wave dipole antenna (or Hertz antenna)
 - Quarter-wave vertical antenna (or Marconi antenna)

3.3.2. Antennae Gain

Is a measure of directionality of an antennae. This is the power of output in particular direction, compared to that produced in any direction by perfect omni direction antennae (isotropic antennae).

- Power output, in a particular direction, compared to that produced in any direction by a perfect omni directional antenna (isotropic antenna)
- Effective area
- Related to physical size and shape of antenna.



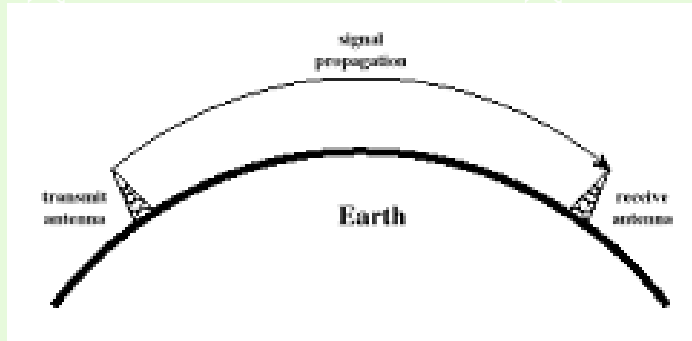
3.3.3. Relationship between antenna gain and effective area

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{C^2}$$

- G = antenna gain
- A_e = effective area
- f = carrier frequency
- c = speed of light (3×10^8 m/s)
- λ = carrier wavelength



3.3.4. Antennae Propagation mode



1. Ground-wave propagation

- Follows contour of the earth
- Can Propagate considerable distances
- Frequencies up to 2 MHz
- Electromagnetic wave induces current in the earth's

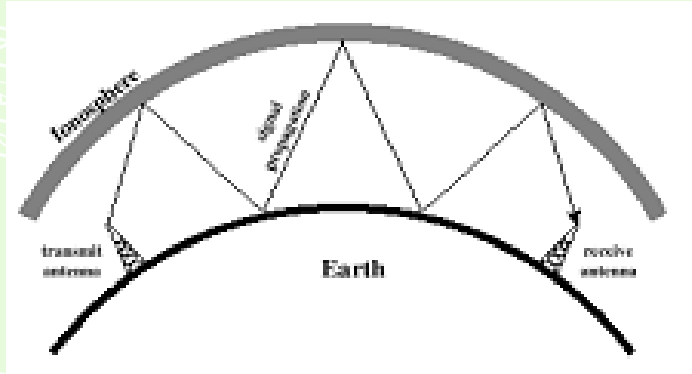


surface, the result of which is to slow the wave front near earth causing the wave front to tilt downwards and hence follow the earth curvature.

- Another factor is diffraction, which is the behavior of electromagnetic signal wave in presence of obstacle.

Example: AM

2. Sky-wave propagation



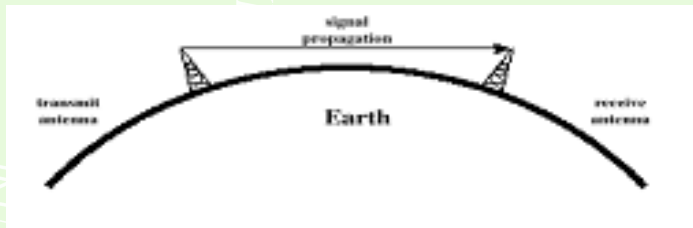
- Signal reflected from ionized layer of atmosphere back down to earth
- Signal can travel a number of hops, back and forth between ionosphere and earth's surface
- Reflection effect is caused by refraction
- A signal can travel number of hops, bouncing back



and fourth between ionosphere and earth's surface.
Signal can travel further.

- Examples: - Amateur radio; - CB radio; - BBC, VOA

3. Line-of-sight propagation



- Transmitting and receiving antennas must be within line of sight
- Satellite communication – signal above 30 MHz not reflected by ionosphere



- Ground communication – antennas within effective line of site due to refraction

• Line-of-Sight Equations

With no intervening obstacles the optical line of sight can be expressed as

- Optical line of sight

$$d = 3.57\sqrt{h}$$

- Effective, or radio, line of sight

$$d = 3.57\sqrt{Kh}$$

- d = distance between antenna and horizon (km)
- h = antenna height (m)
- K = adjustment factor to account for refraction, rule of thumb $K = 4/3$



Example . The maximum distance between two antennas for LOS transmission if one antenna is 100m high and the other is at ground level.

$$D = 3.57\sqrt{133} = 41km$$

3.4. Basic Methods of Propagation

Reflection, diffraction and scattering are the three fundamental phenomena that cause signal propagation in a mobile communication system, apart from LoS communication.

1. Reflection

Reflection occurs when an electromagnetic wave falls on an object, which has very large dimensions as compared



to the wavelength of the propagating wave. For example, such objects can be the earth, buildings and walls. When a radio wave falls on another medium having different electrical properties, a part of it is transmitted into it, while some energy is reflected back.

2. Diffraction

Diffraction is the phenomenon due to which an EM wave can propagate beyond the horizon, around the curved earth's surface and obstructions like tall buildings. As the user moves deeper into the shadowed region, the received field strength decreases. But the diffraction field still exists and it has enough strength to yield a good signal.

3. Scattering

The actual received power at the receiver is somewhat



stronger than claimed by the models of reflection and diffraction. The cause is that the trees, buildings and lampposts scatter energy in all directions. This provides extra energy at the receiver.

3.5. Wireless Transmission Impairments

Any media signal received is different from signal transmitted due to various transmission impairments. For analog this impairments introduces various random modifications that degrade the signal quality. For digital data bit errors are introduced. A binary 1 can be transformed into binary 0 and vice versa.

- Attenuation and attenuation distortion
- Free space loss
- Noise



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- Atmospheric absorption
- Multi-path
- Refraction
- Thermal noise

Mostly, the above impairment affects LOS transmissions.

3.5.1. Attenuation

It is the strength of a signal and falls off with distance over transmission medium

- **Attenuation factors for unguided media:**
 - Received signal must have sufficient strength so that circuitry in the receiver can interpret the signal



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- Signal must maintain a level sufficiently higher than noise to be received without error
- Attenuation is greater at higher frequencies, causing distortion

The first and second factors are dealt with by attention to signal strength by use of amplifiers and repeaters. For point to point links, the signal strength of transmitter must be strong enough to be received intelligibly but not too strong to overload the circuitry which would cause distortion.

- **Free space loss**

For any type of wireless communication, the signal disperses with distance. Therefore any antenna with fixed area will receive less signal power the further it is from the transmitting antenna.



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This is the primary mode of signal loss in satellite. Transmitted signal attenuates over distance because the signal is being spread over a large area.

- Free space loss, ideal isotropic antenna

$$\frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2} = \frac{(4\pi f d)^2}{C^2}$$

- P_t = signal power at transmitting antenna
- P_r = signal power at receiving antenna
- λ = carrier wavelength
- d = propagation distance between antennas
- c = speed of light (3×10^8 m/s)

Where d and λ are in the same units (e.g., meters)



- **Noise**

For any data transmission, the received signal will consist of transmitted signal, modified by various distortions imposed by transmitted signal plus additional unwanted signal. This unwanted signals are referred to as noise.

Categories of Noise

- Thermal Noise
- Intermediation noise
- Crosstalk
- Impulse Noise

1. **Thermal Noise:** This is due to agitation of electrons. It is present in all electronic devices and transmission media.



It is uniformly distributed across the spectrum and is particularly significant for satellite communication. It cannot be eliminated.

Amount of thermal noise to be found in a bandwidth of 1Hz in any device or conductor is:

$$N_0 = kT(W/Hz)$$

- N_0 = noise power density in watts per 1 Hz of bandwidth
- k = Boltzmann's constant = $1.3803 \times 10^{-23} J/K$
- T = temperature, in Kelvin's (absolute temperature)

2. **Intermediation noise:** It occurs if signals with different frequencies share the same medium. Interference caused by a signal produced at a frequency that is the sum or difference of original frequencies



3. **Crosstalk:** This is experienced when you hear another conversation using telephone unwanted coupling between signal paths. Crosstalk can occur if one of the microwave antennas picks unwanted signals.
4. **Impulse noise:** These are irregular pulses or noise spikes of short duration and relatively high amplitude. It can be generated from external electromagnetic disturbances e.g. lightning, faults and flaws in communication systems.
 - Short duration and of relatively high amplitude
 - Caused by external electromagnetic disturbances, or faults and flaws in the communications system



- Other impairments

1. **Atmospheric absorption** – water vapor and oxygen contribute to attenuation

Multi-path –multi-path is the propagation phenomenon that results in radio signals reaching the receiving antenna by two or more paths. Causes of multi-path include atmospheric conducting, ionospheric reflection and refraction, and reflection from water bodies and terrestrial objects such as mountains and buildings. Multiple copies of a signal may arrive at different phases

If phases add destructively, the signal level relative to noise declines, making detection more difficult by receiver difficult.

The effects of multi-path include constructive and destruc-



tive interference, and phase shifting of the signal. This causes Rayleigh fading. The standard statistical model of this gives a distribution known as the Rayleigh distribution.

2. **Refraction** – bending of electromagnetic wave by the atmosphere. Bending occurs because the velocity of electromagnetic wave is a function of the density of the medium through which it travels. In vacuum the electromagnetic wave travels approximately at 3×10^8 m/s and when wave changes medium, speed changes

- Wave bends at the boundary between mediums. From less dense to denser medium, the wave bends towards the denser medium.

3. **Fading**: Fading is deviation or the attenuation that a



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carrier-modulated telecommunication signal experiences over certain propagation media. The fading may vary with time, geographical position and/or radio frequency, and is often modeled as a random process fading in mobile environment can be classified as either fast or slow



Revision Questions

Example . Explain attenuation

Solution:

It is the strength of a signal and falls off with distance over transmission medium □

EXERCISE 1.  List the functions of wireless middleware

EXERCISE 2.  List and explain the categories of noise.



Solutions to Exercises

Exercise 1.

Isolates the application from connectivity issues such as intermittent connections and varying throughput.

Minimizes the amount of data sent over the wireless connection.

Reduces the number of back-and-forth messages required to complete a transaction.

Queues messages when a connection is not available.

Provides a consistent API regardless of the underlying network.

Exercise 1



Exercise 2.

Thermal Noise

Intermediation noise

Crosstalk

Impulse Noise

Explain.

Exercise 2